Collin C. Brandl

Education

University of New Mexico, Albuquerque NM *2020 – 2024*
Ph.D. in Earth and Planetary Sciences
Advisor: Lindsay Worthington
Dissertation: Crustal Structure and Evolution of the Queen Charlotte Plate Boundary
Awards: Outstanding Doctoral Graduate

M.S. in Earth and Planetary Sciences *2018 – 2019*
Thesis: Crustal structure beneath the East Coast Magnetic Anomaly from seismic refraction tomography

Texas A&M University, College Station TX
B.S. in Geophysics, Minors: Physics, Mathematics *2013 – 2017*
Advisor: Bobby Reece
Capstone: Investigating transpressional tectonics near Bonaire with legacy seismic data
Awards: Distinguished Student

Professional Positions

Postdoctoral Research Scientist, Lamont-Doherty Earth Observatory *2024 – Present*
Columbia University (NYC, USA)
Mentor: Anne Bécel

Graduate Teaching and Research Assistant, Dept. of Earth and Planetary Sciences *2018 – 2024*
University of New Mexico, (NM, USA)

Publications

Published:

[5] **Brandl, C. C.,** L. L. Worthington, M. A. L. Walton, B. Phrampus, M. Hornbach, W. Wood, M. Riedel, M. R. Nedimović, E. C. Roland, S. Rysanek, O. Adedeji, J. C. Castellanos, K. Wang (2026), "Subsurface thermal gradients along the Queen Charlotte plate boundary from a regional mapping of bottom simulating reflectors" *Geosphere* – https://doi.org/10.1130/GES02921.1

[4] Van Avendonk, H. J. A., B. D. Shuck, **C. C. Brandl**, T. W. Luckie, A. Bécel, L. L. Worthington, D. J. Shillington, M. B. Magnani, D. Lizarralde (2026), "Limited role for the Central Atlantic Magmatic Province in continental breakup along the eastern United States margin" *Geology* – https://doi.org/10.1130/G54445.1

[3] **Brandl, C. C.,** L. L. Worthington, E. C. Roland, M. Walton, M. R. Nedimović, O. Adedeji, J. C. Castellanos, A. Gase, B. Phrampus, M. Bostock, K. Wang, S. Oliva (2025), "Seismic imaging reveals a strain-partitioned sliver and nascent megathrust at an incipient subduction zone in the northeast Pacific", *Science Advances* – https://doi.org/10.1126/sciadv.adt3003

[2] Gase, A. C., E. C. Roland, L. L. Worthington, M. A. L. Walton, M. G. Bostock, M. R. Nedimović, A. H. Barclay, J. A. Collins, **C. C. Brandl,** S. J. Oliva, T. Sun (2025), "TOQUES: An ocean-bottom seismometer dataset on the central Queen Charlotte Fault, Southeast Alaska", *Seismological Research Letters – Data Mine* – https://doi.org/10.1785/0220240380

[1] **Brandl, C. C.**, L. L. Worthington, M. B. Magnani, D. J. Shillington, T. J. Luckie (2023), "Discontinuous Igneous Addition along the Eastern North American Margin beneath the East Coast Magnetic Anomaly", *Journal of Geophysical Research: Solid Earth* – https://doi.org/10.1029/2023JB026459 – **Editor's Highlight in Eos**

In Review:

[2] **Brandl, C. C.**, A. Bécel, H. Van Avendonk, N. Bangs, E. Parker "Formation of the Blake Plateau from voluminous magmatism during the opening of the central Atlantic" *In review at Earth and Planetary Science Letters*

[1] Rysanek S., L. L. Worthington, M. A. L. Walton, S. Gulick, **C. C. Brandl**, O. A. Adedeji, E. C. Roland, "New constraints on Pleistocene mass flux across the Northern Cordillera from integrated seismic mapping of deep-sea sediment archives" *In review at Geology*

In Prep:

[6] **Brandl, C. C.**, A. Bécel, J. Turney, H. Van Avendonk, N. Bangs, E. Parker "Investigating the disappearance of the East Coast Magnetic Anomaly over the Blake Plateau with geopotential modeling" *To be submitted to Geophysical Journal International*

[5] Parker, E., H. Van Avendonk, N. Bangs, **C. C. Brandl,** A. Bécel, "Building a Magmatic Anomaly: Lithospheric Controls from the Carolina Trough to the Blake Plateau" *To be submitted to Journal of Geophysical Research: Solid Earth*

[4] **Brandl, C. C.,** S. Rysanek, L. L. Worthington, M. A. L. Walton, E. C. Roland, M. R. Nedimović, O. Adedeji, J. C. Castellanos, B. Phrampus, "Widespread Intraplate Magmatism near the Kodiak-Bowie Seamount Chain" *To be submitted to Geophysical Research Letters*

[3] Houghton, E., L. L. Worthington, **C. C. Brandl,** E. C. Roland, A. C. Gase, M. A. L. Walton "Crustal structure of the North American plate along the Queen Charlotte Fault and its potential influence on earthquake mechanics" *To be submitted to Journal of Geophysical Research: Solid Earth*

[2] Adedeji, O.A., L. L. Worthington, M. A. L. Walton, E. C. Roland, D. S. Brothers, M. R. Nedimović, S. Rysanek, **C. C. Brandl**, L. Garza, E. Houghton, A. C. Gase "Strain partitioning and transpressive deformation along the Central to Northern Queen Charlotte Plate Boundary" *To be submitted to Geochemistry, Geophysics, Geosystems*

[1] Rysanek S., L. L. Worthington, M. A. L. Walton, **C. C. Brandl**, O. A. Adedeji, S. Gulick, E. C. Roland, M. R. Nedimović, "Investigating integrated records of tectonic and glacial sedimentary processes in the eastern Gulf of Alaska with multichannel seismic reflection data" *To be submitted to Geochemistry, Geophysics, Geosystems*

Research Proposals and Grants

To be Resubmitted:

National Science Foundation, Marine Geology and Geophysics Full Proposal

"Collaborative Research: Reconstructing the development of forearc slivers during subduction initiation at pre-existing transform plate boundaries"

Award request: $282,456 – Lead PI: Collin Brandl, Co-PIs: Anne Bécel, Paul Betka

- Investigating the formation of forearc slivers with iterative depth migrations and retro-deformation analysis using existing active source seismic data at the Queen Charlotte and Puysegur plate boundaries.

Funded:

International Ocean Discovery Program – U.S. Science Support Program

"Apply-To-Sail: IODP Expedition 385T, Panama Basin Crustal Architecture and Deep Biosphere: Revisiting Holes 504B and 896A" – approximate award: $5,000

Lectures and Seminars

Lamont Doherty Earth Observatory - MPG/SGT Divisional Seminar *March 11th, 2026*
"A Crustal Scale Seismic Investigation of the Blake Plateau"

University of Texas Institute for Geophysics Seminar (*Invited)* *October 3rd, 2025*
"A New Subduction Zone in the Northeast Pacific? Seismic Imaging of the Queen Charlotte plate boundary"

Lamont Doherty Earth Observatory - MPG/SGT Divisional Seminar (*Invited)* *September 25th, 2024*
"A New Subduction Zone in the Northeast Pacific? Oblique Convergence along the Queen Charlotte Plate Boundary"

U.S. Naval Research Lab - Ocean Sciences Division Seminar (*Invited)* *January 18th, 2024*
"Crustal Scale Active Source Imaging of the Southern Queen Charlotte Fault"

Geological Survey of Canada - Pacific Division Seminar (*Invited)* *November 30th, 2023*
"Underthrusting Beneath the Queen Charlotte Terrace Accommodates Recent Plate Convergence"

Guest Lecture for Geological Oceanography (OS211-A) at Maine Maritime Academy *April 13th, 2022*
"Continental Rifting and the Eastern North American Margin (ENAM)"

Rift and Rifted Margins Online Seminar (*Invited)* *November 15th, 2021*
"Along-Strike Discontinuities in Igneous Addition: Evidence for Rift to Ridge Segmentation at the ENAM"

Field Experience

Seismic Experiment at the USGS Albuquerque Seismological Laboratory *August 2023*
Nodal seismometer deployment, sledgehammer seismic survey

Transform Obliquity on the Queen Charlotte Fault and Earthquake Study (TOQUES) *Sept. 2022*
Broadband OBS Recovery aboard R/V *Marcus G. Langseth (10 days at sea)*
Co-Chief Scientists: Lindsay Worthington, Emily Roland
Role: Watch Leader

International Ocean Discovery Program (IODP) Expedition 385T *Aug. – Sept. 2019*
Aboard the JOIDES Resolution *(29 days at sea)*
Co-Chief Scientists: Masako Tominaga, Beth Orcutt
Role: Logging Scientist

Alaska Amphibious Array Community Seismic Experiment (AACSE) *May 2018*
Broadband OBS Deployment aboard the R/V *Sikuliaq (20 days at sea)*
Co-Chief Scientists: Lindsay Worthington, Spahr Webb
Role: Watch Leader

UNOLS Seismic Early Career Chief Scientist Training Cruise *Sept. – Oct. 2017*
Aboard the R/V *Roger Revelle (7 days at sea)*
Chief Scientists: Masako Tominaga, Anne Trehu, Mitch Lyle, Greg Mountain

Teaching Experience

- ENVS (Environmental Science) 102L: The Blue Planet Lab
  Taught for 3 semesters, Supervisor: Dr. Aurora Pun
- ENVS 102: The Blue Planet
  Taught for 1 semester, Supervisor: Dr. Gary Weissman
- EPS (Earth and Planetary Sciences) 400/522: Geophysics for Geology and Environmental Science
  Taught for 1 semester, Supervisor: Dr. Lindsay Worthington

Mentorship

- Sarah Rysanek (Ph.D. Student at University of New Mexico) – seismic reflection processing and interpretation
- Elizabeth Houghton (M.S. Student at University of New Mexico) – refraction tomography, physical properties of the crust
- Davis Hagemeier (Ph.D. Student at Columbia University) – active source seismology, gas hydrate derived thermal gradients/heat flow
- Bailey Fluegel (Undergraduate Student at Northwestern University) – scientific ocean drilling, core-log-seismic integration, active source seismology
- Ingrid Phillips (Undergraduate Student at the University of Washington) – scientific ocean drilling, core-log-seismic integration, active source seismology
- Emma Martin (Undergraduate Student at University of Washington) – active source seismology, high-frequency sub-bottom profiling, multibeam bathymetry processing
- Jacob Tomer (Undergraduate Student at University of Washington) – active source seismology, high-frequency sub-bottom profiling, multibeam bathymetry processing

Academic Service

- Co-organizer of the Blake Plateau/Carolina Trough Active Source Seismology Workshop held at the University of Texas Institute for Geophysics – *January 2026*
- Co-organizer of the LDEO Geodynamics Seminar – *2024 to Present*
- Proposal Reviewer for National Science Foundation – *2025 to Present*
- Peer Reviewer for the Journal of Geophysical Research: Solid Earth, Journal of Geology, New Zealand Journal of Geology and Geophysics – *2024 to Present*
- Presenter at the LDEO Open House, Sound in Water Exhibit – *2024, 2026*
- Co-convener of session at the Seismological Society of America Meeting in Anchorage, Alaska, "Structure, Seismicity and Dynamics of the Queen Charlotte-Fairweather Fault System", Co-conveners: Andrew Gase, Lindsay Worthington, Emily Roland – *2024*
- Co-organizer of the Graduate Student GMT (Generic Mapping Tools) Workshop held at the University of New Mexico, Co-organizer: Han Zhang – *2019*
- Volunteer tour guide for of the Earth and Planetary Sciences' museum at the University of New Mexico – *2018 to 2024*

Conference Abstracts

AGU Fall Meeting, New Orleans, LA, 2025
Collin C. Brandl, Lindsay L. Worthington, Maureen A. L. Walton, Emily C. Roland, Mladen R. Nedimovic, Sarah Rysanek, Olumide Adedeji, Jose Castillo Castellanos, Benjamin J. Phrampus, "Exploring Interactions Between the Bowie Hotspot and the Pacific Plate with Active Source Seismic Data"

AGU Fall Meeting, New Orleans, LA, 2025
Collin C. Brandl, Anne Bécel, Harm Van Avendonk, Nathan Bangs, "Active Source Seismic Data Reveals Syn-Rift Volcanic Variability in the Blake Plateau"

AGU Fall Meeting, New Orleans, LA, 2025
Harm Van Avendonk, Brandon D. Shuck, **Collin C. Brandl**, Thomas W. Luckie, Anne Bécel, Lindsay L. Worthington, Donna J. Shillington, M. Beatrice Magnani, Dan Lizarralde, "CAMP magmatism and continental breakup along the eastern United States margin"

VII Colloquium on Geophysical Signatures of Earthquakes and Volcanoes, October 2025
Sarah J. Oliva, Michael Bostock, Andrew Schaeffer, Ed Nissen, Reid Merrill, Steve Roecker, Alex Hughes, Mladen Nedimović, Emily Roland, Lindsay Worthington, Maureen Walton, Andrew Gase, **Collin Brandl,** Doriane Drolet, "From Transform to Incipient Subduction: Seismotectonic Insights from Haida Gwaii"

AGU Fall Meeting, Washington DC, 2024
Collin C. Brandl, Anne Bécel, Harm Van Avendonk, Nathan Bangs, Sabrina Reichert, "Illuminating the Blake Plateau's Crustal Structure with New Active Source Seismic Data"

AGU Fall Meeting, Washington DC, 2024
Harm Van Avendonk, Nathan Bangs, Anne Bécel, Sabrina Reichert, **Collin C. Brandl,** "New marine seismic data show the extent of syn-rift volcanism during continental breakup offshore the southeastern United States"

AGU Fall Meeting, Washington DC, 2024
Collin C. Brandl, Lindsay L. Worthington, Olumide Adedeji, Jose Castillo Castellanos, Emily C. Roland, Maureen A. L. Walton, Mladen R. Nedimovic, Andrew Gase, Benjamin J. Phrampus, Michael G. Bostock, Kelin Wang, Sarah Jaye Oliva, "Early-Stage Subduction Initiation at an Ocean-Continent Transform Plate Boundary: Crustal Structure of the Queen Charlotte Terrace"

AGU Fall Meeting, Washington DC, 2024
Olumide Adedeji**,** Lindsay L. Worthington, **Collin C. Brandl**, Maureen A. L. Walton, Emily C. Roland, Mladen R. Nedimovic, Lazaro Garza, Andrew Gase, "Slope Evolution and the Accommodation of Oblique Convergence from the Central to the Northern Queen Charlotte Fault"

SSA Annual Meeting, Anchorage, AK, 2023
Collin C. Brandl, Lindsay L. Worthington, Olumide Adedeji, Jose Castillo Castellanos, Emily C. Roland, Maureen A. L. Walton, Mladen R. Nedimovic, Andrew Gase, Benjamin J. Phrampus, "Transpression Along the Southern Queen Charlotte Fault: Underthrusting and Strain Partitioning of the Queen Charlotte Terrace"

AGU Fall Meeting, Chicago, IL, 2022
Collin C. Brandl, Lindsay L. Worthington, Olumide Adedeji, Jose Castillo Castellanos, Emily C. Roland, Maureen A. L. Walton, Mladen R. Nedimovic, Benjamin J. Phrampus, "New Seismic Reflection Imaging of Crustal Deformation in the Queen Charlotte Terrace, offshore Haida Gwaii, Canada"

AGU Fall Meeting, Chicago, IL, 2022
Olumide Adedeji, Lindsay Lowe Worthington, Emily C Roland, Maureen A L Walton, Mladen R Nedimovic, **Collin C. Brandl**, Jose Castillo, Lazaro Garza, Daniel S Brothers, Benjamin J Phrampus, "Strain Partitioning and the Accommodation of Oblique Convergence Along the Northern Queen Charlotte Fault"

AGU Fall Meeting, Chicago, IL, 2022
Lindsay Lowe Worthington, Mladen R Nedimovic, Emily C Roland, United States, Maureen A L Walton, **Collin C. Brand**l, Olumide Adedeji, Lazaro Garza, Jose Castillo, Benjamin J Phrampus, Andrew Gase, Michael G Bostock, Kelin Wang, Canada, Andrew Schaeffer, Tianhaozhe Sun, Canada, Daniel S Brothers, Nathaniel C Miller, "Tectonics and structure offshore the Canadian Cordillera: Insights into strain-partitioning along the Queen Charlotte Fault from comprehensive marine seismic imaging"

AGU Fall Meeting, Chicago, IL, 2022
Emily C Roland, Lindsay Lowe Worthington, Maureen A L Walton, Mladen R Nedimovic, Lazaro Garza, **Collin C. Brandl**, "Crustal structure crossing the Queen Charlotte Fault and Trough in the region of the Haida Gwaii 2012 M7.9 Thrust Earthquake using P-wave tomography"

SEG-AGU Convergent Margins Workshop, Seattle, WA, 2022
Collin C. Brandl, Lindsay L. Worthington, Olumide Adedeji, Jose Castillo Castellanos, Emily C. Roland, Maureen A. L. Walton, Mladen R. Nedimovic, Benjamin J. Phrampus, "New Seismic Reflection Imaging of Crustal Deformation in the Queen Charlotte Terrace, offshore Haida Gwaii, Canada"

GeoPrisms Synthesis Workshop, Honolulu, HI, 2022
Brandl, C. C., L.L. Worthington, M. Beatrice Magnani, Donna J. Shillington, Thomas J. Luckie, "Discontinuous Igneous Addition along the Eastern North American Margin beneath the East Coast Magnetic Anomaly"

GeoPrisms Synthesis Workshop, Honolulu, HI, 2022
L.L. Worthington, Emily C. Roland, Maureen A. L. Walton, Mladen R. Nedimovic, Benjamin J. Phrampus, Olumide Adedeji, **Collin C. Brandl**, Lazaro Garza, and the TOQUES Science Party, "Crustal Deformation, Fault Zone Architecture, and Strain Partitioning on an Ocean-Continental Transform Boundary: New Seismic reflection Imaging on the Queen Charlotte Fault, NE Pacific"

AGU Fall Meeting, New Orleans, LA, 2021
Brandl, C. C., L.L. Worthington, Emily C. Roland, Maureen A. L. Walton, Mladen R. Nedimovic, Benjamin J. Phrampus, Olumide Adedeji, Lazaro Garza, Pinar Gurun, Kevin Pszcola, Sarah Nolt-Caraway, Tianhaozhe Sun, H. Gary Greene, Michael G. Bostock, Kelin Wang, Gwyn Lintern, Andrew Schaeffer (2021), "New Insights on Strain-Partitioning Along the Southern Queen Charlotte Fault, Offshore Haida Gwaii, Canada, from 2021 Multi-Channel Seismic Reflection Imaging"

AGU Fall Meeting, New Orleans, LA, 2021
Adedeji, O., L.L. Worthington, Emily C. Roland, Maureen A. L. Walton, Mladen R. Nedimovic, Benjamin J. Phrampus, **Collin C. Brandl**, Lazaro Garza, Pinar Gurun, Kevin Pszcola, Sarah Nolt-Caraway, Tianhaozhe Sun, H. Gary Greene, Michael G. Bostock, Kelin Wang, Gwyn Lintern, Andrew Schaeffer, "Crustal Deformation and Fault Zone Architecture Along the Queen Charlotte Fault, Offshore Southeast Alaska, Using Long-Offset Multichannel Seismic Data from the Transform Obliquity on the Queen Charlotte fault and Earthquake Study (TOQUES) "

AGU Fall Meeting, New Orleans, LA, 2021
L.L. Worthington, Emily C. Roland, Maureen A. L. Walton, Mladen R. Nedimovic, Benjamin J. Phrampus, Olumide Adedeji, **Collin Brandl**, Lazaro Garza, Pinar Gurun, Kevin Pszcola, Sarah Nolt-Caraway, Tianhaozhe Sun, H. Gary Greene, Michael G. Bostock, Kelin Wang, Gwyn Lintern, Andrew Schaeffer, "Insights into strain- partitioning along a continental-oceanic transform from comprehensive marine seismic imaging of the Queen Charlotte Fault, offshore western Canada and southeast Alaska"

GSA Annual Meeting, Virtual/Portland, OR, 2021
Brandl, C. C., L.L. Worthington, M. Beatrice Magnani, Donna J. Shillington, "Discontinuous High Velocity Lower Crust and Early Margin Segmentation along the Eastern North American Margin's Rift to Drift Transition"

Marine Seismology Symposium, Virtual, 2021
Brandl, C. C., L.L. Worthington, M. Beatrice Magnani, Donna J. Shillington, "Discontinuous High Velocity Lower Crust and Early Margin Segmentation along the Eastern North American Margin's Rift to Drift Transition"

AGU Fall Meeting, Virtual, 2020
Brandl, C. C., L.L. Worthington, M. Beatrice Magnani, Donna J. Shillington, "Discontinuous High Velocity Lower Crust and Early Margin Segmentation along the Eastern North American Margin's Rift to Drift Transition"

Seismological Society of America Annual Meeting, Albuquerque, NM, 2020
Brandl, C. C., L.L. Worthington, M. Beatrice Magnani, Donna J. Shillington, "Along-Strike Crustal Structure of the Eastern North American Margin within the East Coast Magnetic Anomaly"

Student Seismology Workshop, Palisades, NY, 2019
Brandl, C. C., L. Worthington, B. Magnani, B. Shuck, D. Shillington,
"Along-strike magmatic variations of the Eastern North American Margin within the East Coast Magnetic Anomaly**"**

AGU Fall Meeting, Washington D.C, 2018
Brandl, C. C., L. Worthington, B. Magnani, B. Shuck, D. Shillington,
"Along-strike refraction tomography results from the ENAM Community Seismic Experiment within the East Coast Magnetic Anomaly", Abstract T13F-0274

IRIS Workshop, Albuquerque, NM, 2018
Brandl, C. C., L. Worthington, B. Magnani, B. Shuck, H. van Avendonk, D. Shillington, "Initial along-strike refraction tomography results from the ENAM Community Seismic Experiment along the East Coast Magnetic Anomaly"

AGU Virtual Showcase & TAMU Dept. of Geology and Geophysics Symposium, 2017
Brandl, C. C., B. Reece, "Constraining Regional Stress and Deformation around Bonaire and the Bonaire Basin: A Seismic Reflection Study"
AGU Virtual Undergraduate Showcase Award: 3rd Place